

Final Phase Team 5 Contribution

Abanoub Kamel:

He acted as the Embedded Systems and Logic Designer. He focused on the core processing unit, handling the schematic design for the MCU and its essential peripherals, such as the crystal oscillator, reset circuitry, and programming headers. He also managed the integration of the Bluetooth communication module, ensuring proper signal routing for wireless data exchange. To finalize the procurement phase, he was responsible for compiling and verifying the Bill of Materials (BOM), ensuring every component's footprint matched the physical specifications.

John Kamal:

He served as the Systems Architect and Power Electronics Lead. He was responsible for the high-level organization of the project, including the creation of the hierarchical schematic structure and the root page that links all sub-circuits. His technical focus was the design of the power management system, where he implemented the 12V input protection and the dual-stage regulation for the 5V and 3.3V rails. Additionally, he performed the final Electrical Rules Check (ERC) to verify that the global power nets and hierarchical pins were correctly interfaced across the entire design.

Mina Yasser:

He took charge of the Sensor Array and Actuation systems. He designed the infrared sensor schematic using TCRT5000 sensors and implemented the signal conditioning logic through Schmitt triggers to ensure clean data for the MCU. He was also responsible for the motor driver stage, including the selection of the H-bridge controller and the layout of the flyback diode protection array. Beyond the hardware, he was the primary author of the Finite State Machine (FSM) diagram, defining the logical behavior and operational states of the system's firmware.

Omar Elshazli:

He functioned as the PCB Layout and Manufacturing Engineer. His primary task was the physical realization of the circuit, which involved placing all components and routing the traces on the multi-layer board. He specifically focused on signal integrity by managing copper pours for ground planes and ensuring the Bluetooth antenna area remained clear of interference. Following the layout, he executed the Design Rules Check (DRC) to ensure the board met manufacturing tolerances and prepared the final Gerber files and 3D visualizations required for the production phase.